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ABSTRACT

Your abstract text goes here. Keep it concise, focused on the core database or data management challenge, your approach, and key experimental findings.

CCS CONCEPTS

• Information systems → Data management systems.

KEYWORDS

databases, query processing, distributed systems

1 INTRODUCTION

2 RELATED WORK

3 METHODOLOGY

3.1 Preliminaries

3.2 Query Delegation in Discovery Settings

Dynamically integrating conjunctive sub-queries during federated query processing in discovery-based querying at runtime is difficult due to the unbounded nature of the query execution. To show this, we first introduce a sufficient condition for correctly delegating conjunctive queries in the traditional federated case. Then we will show how deciding whether we can correctly delegate these queries in the discovery-based querying case is impossible when not all sources have been discovered.

First we define our discovery-based federated query setting in Definition 1 and 2.

Definition 1 (Discovery setting). A *discovery setting* is a tuple $W = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc} \rangle$: a universe of sources $\mathbb{D}$, a map $\text{data} : \mathbb{D} \rightarrow 2^T$ giving the graph each one holds, a set of seeds $\mathcal{D}_0 \subseteq \mathbb{D}$, and a *discovery function* disc assigning to every BGP B a map $\text{disc}_B : \mathbb{D} \times 2^T \rightarrow 2^{\mathbb{D}}$, where $\text{disc}_B(d, \text{data}(d))$ are the sources that a retrieved d exposes while B is evaluated.

Definition 2 (Reachable set, reference answer). Iterating discovery from the seeds, let $\mathcal{D}_{i+1} := \mathcal{D}_i \cup \bigcup_{d \in \mathcal{D}_i} \text{disc}_B(d, \text{data}(d))$. The *reachable set* is $\text{reach}_B(W) := \bigcup_{i \geq 0} \mathcal{D}_i$, and the *reference answer* to B is $\llbracket B \rrbracket_{G_{\text{reach}_B(W)}}$. Correctness of a delegation is always measured against this reference answer, never against the universe $\mathbb{D}$: sources that are never reached cannot contribute to it.

Given this definition, we recover the traditional federated query processing setting, in which all sources are known beforehand, by taking $\mathcal{D}_0 = \mathbb{D}$ and $\text{disc}_B \equiv \emptyset$.

Next we define an operator to determine the *relevance* of a source for a query execution setting. Relevance records, for each triple pattern and each source, whether that source produces matches. An example of such an operator is the usage of ASK queries to determine exclusive groups in FedX [2].

Definition 3 (Source relevance). Fix a setting W and a BGP B , and write $\mathcal{D} := \text{reach}_B(W)$ for its reachable set; all that follows is

relative to $\mathcal{D}$. Let $\mathcal{P}$ be the set of triple patterns. We write $\llbracket t \rrbracket_d$ for the evaluation of a triple pattern t over the data of source d . The *relevance indicator* $\Delta : \mathcal{P} \times \mathcal{D} \rightarrow \{0, 1\}$ is

$$\Delta(t, d) = \begin{cases} 1, & \text{if } \llbracket t \rrbracket_d \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases}$$

and $\mathcal{D}_t := \{d \in \mathcal{D} \mid \Delta(t, d) = 1\}$ denotes the *relevant sources* of t .

Thus, relevance means that a data source returns answers for a given triple pattern. Next, we introduce a definition of *correct* delegation of conjunctive queries to a set of sources.

Definition 4 (Delegation). Let B be a BGP, $B' \subseteq B$ the delegated fragment, and $S \subseteq \mathcal{D}$ the (only) sources to which B' is sent as a single conjunctive request, where as in Definition 3, $\mathcal{D} = \text{reach}_B(W)$. Write $G_A := \bigcup_{d \in A} \text{data}(d)$ for $A \subseteq \mathcal{D}$. The delegated plan obtains for B'

$$R := \left(\bigcup_{d \in S} \llbracket B' \rrbracket_{\text{data}(d)} \right) \cup \llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}},$$

and the delegation is *correct* if

$$R \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}} = \llbracket B \rrbracket_{G_{\mathcal{D}}}.$$

This definition follows some of the definitions of federation of FedQPL [1]. Next we give a sufficient condition for correctly delegating conjunctive (sub)-queries to federated sources.

Theorem 1 (Correct delegation). Let W be a federated setting, that is $\text{disc}_B \equiv \emptyset$, so that $\mathcal{D} = \text{reach}_B(W) = \mathcal{D}_0$ is known in advance. Let $t_1, \dots, t_n$ be a set of triple patterns forming a BGP B , $B' \subseteq B$ a sub-BGP, and let $d \in \mathcal{D}$. For d we define $\mathcal{D}' = \mathcal{D} \setminus \{d\}$ the set containing all other sources. Then we can correctly delegate B' to d if $\bigcup_{t_i \in B'} \{d' \in \mathcal{D}' \mid \Delta(t_i, d') = 1\} = \emptyset$

PROOF. We prove the special case of Definition 4 with $S = \{d\}$, so that $\mathcal{D} \setminus S = \mathcal{D}'$. We assume $B' \neq \emptyset$ and that distinct sources use disjoint sets of blank nodes.

The argument rests on the fact that a *single* triple pattern distributes over a union of sources: for every t and every $A \subseteq \mathcal{D}$,

$$\llbracket t \rrbracket_{G_A} = \bigcup_{e \in A} \llbracket t \rrbracket_{\text{data}(e)}, \quad (1)$$

since matching one pattern involves no join. Note that (1) fails for BGPs, which is exactly what makes delegation non-trivial.

By hypothesis no source other than d is relevant to any pattern of B' , so Definition 3 gives

$$\llbracket t' \rrbracket_{\text{data}(d')} = \emptyset \quad \text{for all } t' \in B', d' \in \mathcal{D}'. \quad (2)$$

Step 1. A BGP is the join of its patterns, and each pattern distributes over the sources by (1):

$$\llbracket B \rrbracket_{G_{\mathcal{D}}} = \bowtie_{t \in B} \llbracket t \rrbracket_{G_{\mathcal{D}}} = \bowtie_{t \in B} \bigcup_{e \in \mathcal{D}} \llbracket t \rrbracket_{\text{data}(e)}.$$

Step 2. Since $\bowtie$ is associative and commutative, the delegated fragment splits off:

$$\llbracket B \rrbracket_{G_{\mathcal{D}}} = \left(\bowtie_{t' \in B'} \bigcup_{e \in \mathcal{D}} \llbracket t' \rrbracket_{\text{data}(e)} \right) \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}}.$$

Step 3. As $\mathcal{D} = \mathcal{D}' \uplus \{d\}$, the inner union splits, and by (2) its $\mathcal{D}'$ part is empty:

$$\bigcup_{e \in \mathcal{D}} \llbracket t' \rrbracket_{\text{data}(e)} = \underbrace{\bigcup_{d' \in \mathcal{D}'} \llbracket t' \rrbracket_{\text{data}(d')} \cup \llbracket t' \rrbracket_{\text{data}(d)}}_{= \emptyset} = \llbracket t' \rrbracket_{\text{data}(d)}.$$

Substituting into Step 2 and reassembling the join,

$$\llbracket B \rrbracket_{G_{\mathcal{D}}} = \llbracket B' \rrbracket_{\text{data}(d)} \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}}. \quad (3)$$

Step 4. It remains to identify R from Definition 4. Its second term vanishes, again by (1) and (2) and using $B' \neq \emptyset$:

$$\llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}} = \bigwedge_{t' \in B'} \bigcup_{d' \in \mathcal{D}'} \llbracket t' \rrbracket_{\text{data}(d')} = \bigwedge_{t' \in B'} \emptyset = \emptyset.$$

With $S = \{d\}$ the first term of R is the single request to d , so $R = \llbracket B' \rrbracket_{\text{data}(d)} \cup \emptyset = \llbracket B' \rrbracket_{\text{data}(d)}$.

Conclusion. Substituting R into (3),

$$R \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}} = \llbracket B \rrbracket_{G_{\mathcal{D}}},$$

which is the correctness condition of Definition 4. $\square$

Theorem 1 gives us a sufficient condition for delegating conjunctive sub-queries in the case where the reachable set $\mathcal{D}$ is known in advance. However, in discovery-based querying the sources are discovered at runtime, governed by some discovery process which returns during the query execution. As such, determining if we can correctly delegate depends on the *state* of the discovery process, which is defined in Definition 5.

Definition 5 (Discovery state). The state of a discovery setting $W = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc} \rangle$ at time T is the tuple

$$W(T) = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc}, \mathcal{D}(T) \rangle,$$

the setting W augmented with $\mathcal{D}(T)$, the set of sources that data has been called on by time T , subject to

$$\mathcal{D}_0 \subseteq \mathcal{D}(T) \subseteq \mathcal{D}(T') \subseteq \text{reach}_B(W) \quad \text{for all } T \leq T',$$

i.e. the seeds are discovered from the start, discovery only ever adds sources, and it never adds one outside what disc can reach. For any given $W(T)$ the discovery *frontier* is

$$F(T) = \left(\bigcup_{d_f \in \mathcal{D}(T)} \text{disc}(d_f, \text{data}(d_f)) \right) \setminus \mathcal{D}(T),$$

the sources that are discovered but do not yet have data called on them. Discovery is complete when $F(T) = \emptyset$.

Next, we introduce the concept of an observation of an engine of the discovery state in Definition 6, which denotes what the engine actually can *know* about its discovery setting at given time T .

Definition 6 (Observationally equivalent states). A discovery state $W(T)$ records the whole world, but a query engine at time T has access only to the *observation*

$$O(T) = \langle \mathcal{D}(T), \text{data}|_{\mathcal{D}(T)} \rangle,$$

the sources it has retrieved together with their contents. Two states $W_1(T)$ and $W_2(T)$ are *observationally equivalent*, written $W_1(T) \sim W_2(T)$, when $O_1(T) = O_2(T)$. Note that $F(T)$ is computable from $O(T)$ and disc: the extent of the frontier is observable, its contents are not. A property of states is *determined at T* when all states

observationally equivalent to $W(T)$ agree on it; an engine at time T can establish a property only if it is determined at T .

Theorem 2 (Correct delegation is undetermined under incomplete discovery). Let $W(T)$ be a discovery state for which $F(T) \neq \emptyset$. Then there exist a BGP B , a sub-BGP $B' \subseteq B$, a set of sources $S \subseteq \mathcal{D}(T)$, and two states $W_1(T)$ and $W_2(T)$, both observationally equivalent to $W(T)$, such that B' is correctly delegated to S in $W_1(T)$ but not in $W_2(T)$. Since the engine's input is $O(T)$ and all three states share it, no procedure can decide whether the delegation is correct.

PROOF. We construct two states $W_1(T)$ and $W_2(T)$, both observationally equivalent to $W(T)$, in which B' is delegated to S correctly and incorrectly respectively. The two agree on $\mathcal{D}_0$, disc and $\mathcal{D}(T)$, and on data over $\mathcal{D}(T)$; they differ only beyond the frontier, which is non-empty by hypothesis.

Take $\mathcal{D}(T) = S = \{d_1\}$ and

$$B' = B = \{ (?s, p_1, ?o_1), (?s, p_2, ?o_2) \},$$

$$\text{data}(d_1) = \{ (s_1, p_1, o_1), (s_1, p_2, o_2), (s_2, p_1, o_3) \}.$$

Fix some $d^* \in F(T)$, and let every reachable source other than d_1 and d^* hold no data. The two states are then obtained by setting

$$\text{data}(d^*) = \begin{cases} \{ (s_3, p_2, o_4) \} & \text{in } W_1(T), \\ \{ (s_2, p_2, o_4) \} & \text{in } W_2(T). \end{cases}$$

Since $d^* \notin \mathcal{D}(T)$, both states induce the same observation, and so $W_1(T) \sim W_2(T) \sim W(T)$ holds. The reachable set differs between the two, so we write G_i for $G_{\text{reach}_B(W_i)}$, the union of all data reachable in $W_i(T)$. As only d_1 and d^* hold data,

$$G_1 = \text{data}(d_1) \cup \{ (s_3, p_2, o_4) \}, \quad G_2 = \text{data}(d_1) \cup \{ (s_2, p_2, o_4) \}.$$

Write

$$\mu_1 = \{ ?s \mapsto s_1, ?o_1 \mapsto o_1, ?o_2 \mapsto o_2 \},$$

$$\mu_2 = \{ ?s \mapsto s_2, ?o_1 \mapsto o_3, ?o_2 \mapsto o_4 \}.$$

Neither added triple matches $(?s, p_1, ?o_1)$, so $\llbracket B' \rrbracket_{\text{data}(d^*)} = \emptyset$ in both states. By Definition 4 we therefore have

$$R_1 = R_2 = \llbracket B' \rrbracket_{\text{data}(d_1)} = \{ \mu_1 \}.$$

Since $B \setminus B' = \emptyset$ we have $\llbracket B \setminus B' \rrbracket_{G_i} = \{ \mu_0 \}$, the join identity, so the correctness condition reduces to $R_i = \llbracket B \rrbracket_{G_i}$.

In $W_1(T)$ the added triple has subject s_3 , which no p_1 -triple mentions, so it contributes no solution and

$$\llbracket B \rrbracket_{G_1} = \{ \mu_1 \} = R_1 :$$

the delegation is correct. In $W_2(T)$ the added triple joins with $(s_2, p_1, o_3) \in \text{data}(d_1)$ on $?s$, so

$$\llbracket B \rrbracket_{G_2} = \{ \mu_1, \mu_2 \} \neq R_2 :$$

the delegation is not correct.

The triple at d^* yields no solution on its own, and (s_2, p_1, o_3) yields none within d_1 ; only their join does. A single conjunctive request to d_1 cannot produce it, and neither can one to d^* , so the solution is lost precisely because the join crosses the frontier. As $W_1(T)$ and $W_2(T)$ present the engine with the same observation $O(T)$ while disagreeing on correctness, no procedure reading only $O(T)$ can decide it. $\square$

Lemma 1 (An empty frontier certifies complete discovery). $F(T) = \emptyset$ if and only if $\mathcal{D}(T) = \text{reach}_B(W)$.

PROOF. ($\Rightarrow$) The inclusion $\mathcal{D}(T) \subseteq \text{reach}_B(W)$ holds by Definition 5. For the converse, $F(T) = \emptyset$ says exactly that

$$\text{disc}(d_f, \text{data}(d_f)) \subseteq \mathcal{D}(T) \quad \text{for every } d_f \in \mathcal{D}(T),$$

i.e. that $\mathcal{D}(T)$ is closed under disc ; and $\mathcal{D}_0 \subseteq \mathcal{D}(T)$, again by Definition 5. By induction, if $\mathcal{D}_i \subseteq \mathcal{D}(T)$ then

$$\mathcal{D}_{i+1} = \mathcal{D}_i \cup \bigcup_{d \in \mathcal{D}_i} \text{disc}(d, \text{data}(d)) \subseteq \mathcal{D}(T)$$

by closure, so $\text{reach}_B(W) = \bigcup_{i \geq 0} \mathcal{D}_i \subseteq \mathcal{D}(T)$.

($\Leftarrow$) If $d \in \text{reach}_B(W)$ then $d \in \mathcal{D}_i$ for some i , hence

$$\text{disc}(d, \text{data}(d)) \subseteq \mathcal{D}_{i+1} \subseteq \text{reach}_B(W).$$

Thus $\text{reach}_B(W)$ is closed under disc , and $F(T) = \emptyset$ follows. $\square$

{TODO: Don't like the text here} is what makes the frontier useful: $F(T) = \emptyset$ is a condition the engine can test, whereas $\mathcal{D}(T) = \text{reach}_B(W)$ is a statement about a set it cannot inspect. Together with Theorem 2 it fixes exactly when delegation can be established at all.

Corollary 1 (Delegation requires complete discovery). The correctness of every delegation of a sub-BGP $B' \subseteq B$ to a set $S \subseteq \mathcal{D}(T)$ is determined at T if and only if $F(T) = \emptyset$.

PROOF. ($\Leftarrow$) Let $F(T) = \emptyset$ and let $W_1(T), W_2(T)$ be observationally equivalent to $W(T)$. As $F(T)$ is computable from $O(T)$, both have an empty frontier, so $\mathcal{D} = \text{reach}_B(W) = \mathcal{D}(T)$ in both by Lemma 1. Then $G_{\mathcal{D}} = \bigcup_{e \in \mathcal{D}(T)} \text{data}(e)$ agrees in both, and since $S \subseteq \mathcal{D}(T) = \mathcal{D}$, so do $G_{\mathcal{D} \setminus S}$ and $\text{data}(d)$ at each $d \in S$. Every term of the condition of Definition 4 therefore agrees, and with it the verdict.

($\Rightarrow$) By contraposition. If $F(T) \neq \emptyset$, then Theorem 2 supplies a B' , an S , and two states observationally equivalent to $W(T)$ that disagree on the correctness of delegating B' to S . That delegation is not determined at T . $\square$

3.3 Authority Claims

From Corollary 1, runtime federation in discovery-based querying is impossible until every source is discovered. With source counts in the thousands **{TODO: cite here}**, this is infeasible.

However, while classical query semantics require all data to be considered in query execution, this is not always an intuitive requirement. Consider a federation of product catalogues, in which a manufacturer publishes the specifications of the products it makes while resellers and comparison services republish copies of them. A query over a product's specification is answered from the manufacturer alone. However, if any copies of, different, or wrong specifications exist in the federation, they are considered equally 'true'.

We will work towards a query execution model that resolves this observation while also allowing the delegation of queries *during* query execution. First we introduce the concept of a *composite source*, an (aggregating) view over sources in $\mathbb{D}$ which contains no data of its own.

Definition 7 (Composite source). Fix a setting $W = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc} \rangle$. A source $c \in \mathbb{D}$ is *composite* when it holds no data of its own and is instead given by a pair $\langle \mathcal{D}_c, \sigma_c \rangle$, with $\mathcal{D}_c \subseteq \mathbb{D}$ a fixed set of sources it aggregates and $\sigma_c : 2^{\mathcal{T}} \rightarrow 2^{\mathcal{T}}$ a *selector* satisfying $\sigma_c(G) \subseteq G$ for every $G \subseteq \mathcal{T}$ and given by a triple-wise decidable description. Its data is then

$$\text{data}(c) = \sigma_c \left(\bigcup_{d \in \mathcal{D}_c} \text{data}(d) \right).$$

Sources that are not composite are *base* sources. We require the aggregation relation to be well-founded: no source aggregates itself, whether directly or through a chain of other composite sources. Unfolding the definition of $\text{data}(c)$ then terminates at base sources, which is what makes data a well-defined map on all of $\mathbb{D}$. In the degenerate case $\mathcal{D}_c = \{d\}$ with d a base source we obtain $\text{data}(c) = \sigma_c(\text{data}(d))$, a filtered view of a single source, and with σ_c the identity a plain mirror of it.

The key assumption to facilitate the delegation under incomplete discovery is to allow composite sources to *declare* that it is authoritative over a part of the data in a discovery setting that it aggregates over and that an engine may take that declaration at face value. In the product catalogue federation case, a manufacturer can declare that only it can make statements about their products. This declaration depends on a *authoritativeness* decider function as seen in Definition 8.

Definition 8 (Authoritativeness declaration). Take a setting W with data function data . A composite source c may *declare* authority over its selector σ_c , meaning that the triples σ_c selects are to be taken from c and from no other source. Write $C \subseteq \mathbb{D}$ for the composite sources that declare, and put $\sigma_d = \emptyset$ for any source $d \notin C$. The *authoritative data* of a source is then what it may still speak for,

$$\theta(d, G) = \{ t \in G \mid t \in \sigma_d(G) \text{ or } t \notin \bigcup_{c \in C} \sigma_c(G) \},$$

so that a triple survives at d when d itself declares it, or when no source declares it at all. Evaluation is carried out over authoritative data rather than stored data: for $A \subseteq \mathbb{D}$ we replace G_A of Definition 4 by

$$G_A = \bigcup_{d \in A} \theta(d, \text{data}(d)).$$

Thus, when a source claims *authority* over a part of the discovery setting, *completeness* of the query result is now with respect to the new G_A .

{TODO: Tie to literature here about subwebs and relational}
{TODO: Paragraph about scope of claim and anchor mapping definition} **{TODO: We cannot allow anyone to say anything about another source. We must determine who can say what}** Nothing so far prevents a source from selecting data it has no business aggregating. We therefore define an anchor mapping, which determines whether a declared σ_c is valid.

Definition 9 (Anchor mapping).

{TODO: This part comes after the definitions this is where the first real function definition comes. Based on this we will prove we can determine } We limit a declaration to triples whose *subject* falls within the declared scope. The picture is that of horizontal

partitioning, with each participant the sole source of record for a region of the key space. The difference is that the partitioning is declared by the participants rather than imposed by a coordinator, so no global catalogue of it exists and both the participants and the partitioning are discovered during execution. **{TODO: Two drafts}** In this work, we will use a specific form of this mapping, which aligns with **{TODO: Subweb paper here}**.

Definition 10.

3.4 Delegating Conjunctive Queries

Under this query execution model, we can now investigate delegating queries to composite sources when our model allows determining *correct* delegation under *incomplete* discovery.

{TODO: Lemma on how theorem 1 holds if we filter any sources in scope of c both backwards and forwards in time. Mention how the filtering changes nothing given the assumptions / semantics of our model}

PROOF. Sketch Part 1. If we filter $\rightarrow$ Theorem 1 can be directly applied Part 2. Filtering is possible: forward $\rightarrow$ easy just take `disc()` and apply `setminus` of in scope documents. Backwards $\rightarrow$ deduplication, we only have duplicate results. So filter any intermediate results at time T in `data(scope)` and filter any produced results in composite resource on B prime $\square$

3.5 Adaptive Query Planning with Composite Sources

Given the capability of delegating these sub queries, we must now move to the system-side of our problem. Specifically, due to the dynamic discovery of composite sources and the uncertainty of capabilities of these sources, using a static query plan is difficult.

{TODO: Example showing how we can't integrate a sub-query if we have a static plan}

{TODO: Show that even if we use many alternative plans we can still run into issues with overlapping composite sources}

{TODO: Stems!}

3.6 Correctness

3.6.1 Overlapping Sources. Correctness when doing overlapping sources is difficult blabla, cases: Overlapping domain, overlapping triple patterns, both, neither. Given the time dimension, how do we decide what is possible

3.7 Discovery

4 SYSTEM APPLICATION

To evaluate the performance impact of delegating sub-queries we implement hybrid federated link traversal-based query processing. Specifically, within solid blabla

Note this is a concrete application of the stuff introduced here, which is not limited to Solid and link traversal.

5 EVALUATION

TODO: Ensure correctness of multiple composite resources!! Experiments: - Full benchmark and abalations - Client-scaling test (find server-side usage and client-side scaling behavior) - Query where

linear is fastest to delegate with single pod (show star-only fails and cost-based fails) - Query where linear is fastest to delegate with multiple pods (show if cost-based doesn't fail) - Query where star is fastest multiple pods (show if cost-based doesn't fail) - Multiple vs Single pod query

6 CONCLUSION

Summarize results and point to future work.

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